

Generalization of Activation-Based Safety Monitors

Probe coverage, transferable representations,
and prompt interventions

Working research notes for Pranav Konda

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Status: Proposed definitions and experiments, prepared from the application discussion. No experiments were run for this note. The mathematical observations are elementary; neither novelty nor a general safety guarantee is claimed.

1 The common question and measurement contract

Question. Which behaviorally consequential distinctions remain invisible to a collection of internal monitors, and what additional measurement or intervention addresses the failure?

Three hypotheses organize the project:

1. Different probes may have complementary errors, so their combination improves held-out detection.
2. Some activation features may preserve behavior-relevant distinctions across changes of task, interface, and violation mechanism.
3. Prompting may change those features in a way that reduces independently verified violations while preserving useful behavior.

These are separate claims: prediction, representation sufficiency, and causal control require different evidence.

Fix the actor model, monitoring time, and activation access. Let $e \in \mathcal{E}$ index an environment, such as a task family or violation mechanism; $H \in \mathbb{R}^d$ denote observed activations; C denote declared task information available to the monitor; $Y \in \{0, 1\}$ indicate an independently verified violation; and U measure useful completion. Record whether the monitor observes a proposed action, a completed action, or only a state preceding future behavior. Do not mix these targets.

A score $m(H, C)$ raises an alarm at a frozen threshold t . Define

$$\text{FNR}_e(m, t) = \mathbb{P}_e(m(H, C) < t \mid Y = 1), \quad (1)$$

$$\text{FPR}_e(m, t) = \mathbb{P}_e(m(H, C) \geq t \mid Y = 0). \quad (2)$$

Report worst-environment FNR and FPR as well as individual environments. Choose methods and thresholds using development data; evaluate held-out environments without recalibration. AUROC is a useful descriptive statistic, but does not replace an operational false-positive budget. Rates require enough independent positive and negative cases; model calls are not independent task replications.

Scope. A verified violation is not automatically deception, reward-seeking, or a hidden intention. Define these separately if they become research targets. For out-of-distribution evaluation, withhold an entire mechanism or task family, not merely another paraphrase of a familiar instance.

Motivation from prior work. Anthropic’s sleeper-agent probes achieved AUROC above 0.99, while explicitly noting that their construction may make defection unusually salient internally [1]. Its later honesty study found that truth probes and honesty steering underperformed simpler prompting and fine-tuning approaches in the studied testbeds [2]. The user’s source-reported small-model pilot similarly found candidate AUROC 1.000/1.000 falling to 0.590/0.525 on natural stress actions. This motivates a transfer question; it does not replicate or refute the Anthropic experiments.

2 Hypothesis 1: complementary probes cover shared blind spots

Let $p_j(H, C)$ be the score of probe j , and collect scores into

$$P(H, C) = (p_1(H, C), \dots, p_k(H, C)).$$

Compare an OR rule, a calibrated weighted average, and a small learned combiner $g(P, C)$. For the OR rule, writing $F_j = \{p_j < t_j\}$,

$$\text{FNR}_{\text{OR}} = \mathbb{P} \left(\bigcap_{j=1}^k F_j \mid Y = 1 \right), \quad \text{FPR}_{\text{OR}} \leq \sum_{j=1}^k \text{FPR}_j. \quad (3)$$

Thus the relevant diversity is *non-overlapping misses on violations*. Do not multiply error rates without an independence assumption. An ensemble can appear more sensitive simply because it generates more false alarms; compare at the same total false-positive budget and comparable inference cost.

2.1 A local geometric diagnostic

For intervention experiments, use a context c specifying the complete recipient continuation outside the patched coordinates, including relevant cache state. This is richer than the monitor’s public context C . Let h denote coordinates at a fixed patch site and $\mathcal{A}_c$ an explicitly specified admissible intervention set. Define

$$B_c(h) = \mathbb{P}(\text{verified violation after intervention } h \text{ in context } c).$$

Use repeated continuations to estimate this probability when decoding is stochastic. A proxy such as a report-logit margin is a different outcome and must be labeled accordingly.

On a smooth admissible patch family, a first-order shared blind direction satisfies

$$DP_c(h)v = 0, \quad DB_c(h)v \neq 0.$$

The desired first-order coverage condition is

$$\boxed{\ker DP_c(h) \subseteq \ker DB_c(h)}, \quad (4)$$

with both derivatives restricted to the allowed tangent directions. This says that no infinitesimal behavior-relevant change is invisible to the whole bank. It does not establish global sufficiency: higher-order effects, poor conditioning, or a lossy learned combiner may still matter. If the admissible family is discrete, use finite comparisons instead of assuming a tangent manifold.

2.2 A finite-change target

Define the behavior difference concealed by nearly identical probe readings:

$$\omega_{P,c}(\delta) = \sup_{\substack{h, h' \in \mathcal{A}_c \\ \|P_c(h) - P_c(h')\| \leq \delta}} |B_c(h) - B_c(h')|. \quad (5)$$

Choose and freeze score scaling before interpreting the tolerance δ . If $P_c(h) = P_c(h')$ but the two behavioral risks differ by Δ , every risk predictor using only these scores has absolute error at least $\Delta/2$ on one state. This follows immediately from the triangle inequality. A nonlinear combiner cannot recover an exactly discarded distinction.

An observed witness lower-bounds ω ; an unsuccessful finite search does not upper-bound it. This formulation extends the earlier task-relative observability memo from a report channel to a bank of probe outputs.

First test. Freeze the bank and thresholds, evaluate shared errors on a withheld mechanism, and search for matched cases or admissible interventions with similar readings but different verified outcomes. Select any additional probe on discovery cases and test whether it resolves *new* witnesses. Stop the ensemble-improvement claim if gains disappear at matched FPR or are explained by more compute.

3 Hypothesis 2: a representation preserves transferable distinctions

Replace an assumed “ideal misalignment representation” with a narrower object: a representation sufficient for a declared behavioral target across a declared environment family.

Let $Z = \phi(H)$. A strong predictive hypothesis is that a common readout satisfies

$$\boxed{\mathbb{P}_e(Y = 1 \mid Z = z, C = c) = r(z, c) \text{ for all tested } e,} \quad (6)$$

on relevant overlapping support. Environments may change the distribution of Z while preserving its relation to behavior. This connects to invariant prediction [3]; it is not a guarantee of generalization to arbitrary unseen environments. Disjoint supports can make a fitted common readout appear invariant without a meaningful test.

Construct two kinds of paired cases.

- *Behavior-preserving pairs:* change names, formatting, or presentation while retaining the same candidate action and verified validity.
- *Behavior-changing contrasts:* keep the proposed action fixed while changing permission or protected ground truth so that its validity changes.

The combined representation and readout should ignore irrelevant changes while retaining the information needed for the second distinction. If the changed permission is supplied through C , the readout must use it appropriately; not every semantic distinction must reside in Z alone.

For appropriately matched nuisance pairs, one candidate penalty is

$$\mathcal{L}_{\text{nuisance}} = \mathbb{E} \|\phi(H) - \phi(H')\|^2.$$

Combine this with prediction and validity-changing contrast objectives. A constant representation minimizes the nuisance penalty and is useless. Furthermore, a semantics-preserving prompt change can

alter the actor’s behavior; check outcomes rather than assuming generated trajectories have the same label.

A stronger causal version. For a fixed family $\mathcal{I}$ of continuation operations or downstream interventions, ask whether

$$B_c(h, a) \approx r(\phi(h), c, a), \quad a \in \mathcal{I}. \quad (7)$$

States with similar representations should have similar response profiles under the same operations. Observational classification alone does not establish this property. Use matched interventions, independent outcome measurements, and a declared intervention domain.

First test. Fit a direction or low-dimensional subspace on several mechanisms, freeze it, and evaluate prediction and response-profile transfer on a withheld mechanism. Compare a conventional linear probe, a nonlinear reader, and a strong text monitor. Distinguish information absent from the chosen representation from information missed by one reader. Attractive clusters or a successful linear fit do not establish a universal concept.

4 Hypothesis 3: prompt interventions improve actual behavior

Randomize a prompt intervention u over matched underlying tasks. Write (H^0, Y^0, U^0) for the baseline and (H^u, Y^u, U^u) for the intervention. Keep the monitor fixed and measure

$$\Delta_Y(u) = \mathbb{E}[Y^u - Y^0], \quad \Delta_m(u) = \mathbb{E}[m(H^u, C) - m(H^0, C)]. \quad (8)$$

These quantities answer different questions:

Observation	Interpretation to investigate
Risk and score both fall	Consistent with a useful intervention; inspect usefulness and transfer.
Score falls, risk does not	The monitor reading changes without behavioral improvement.
Risk falls, score does not	The monitor misses a useful behavioral change.
Risk falls, usefulness collapses	Refusal or incapacity may explain the apparent safety gain.

A development objective for selecting prompts is

$$\min_{u \in \mathcal{U}} \max_{e \in \mathcal{E}_{\text{dev}}} \mathbb{E}[Y^u \mid e], \quad (9)$$

$$\text{subject to } \mathbb{E}[U^u \mid e] \geq \mathbb{E}[U^0 \mid e] - \eta \quad \text{for all } e \in \mathcal{E}_{\text{dev}}. \quad (10)$$

Freeze the selected prompt and evaluate it on untouched mechanisms. Anthropic’s persona-vector work provides related examples of distinguishing activation measurements, inference-time steering, and interventions during training [4].

Test a proposed mechanism separately. Transfer a selected component from the prompted run into the original recipient continuation; measure whether this reproduces part of the behavioral improvement. Include matched random patches, irrelevant donors, and removal/rescue controls. Preserve the recipient’s other continuation state explicitly. A patch effect establishes a causal effect of that patch in the tested setup, not complete mediation of the prompt effect: prompts can alter many positions, attention patterns, and cached states.

5 Geometric separation: start with a safety margin

A claim that safe and unsafe states are “geometrically disconnected” requires a specified state space and allowed paths. A scatterplot does not establish it. Begin with a robustness quantity instead.

For the same fixed continuation context and admissible set used above, define

$$\mathcal{S}_{\epsilon,c} = \{h \in \mathcal{A}_c : B_c(h) \leq \epsilon\}.$$

For $h \in \mathcal{S}_{\epsilon,c}$, its distance to a higher-risk state is

$$\rho_{\epsilon,c}(h) = \inf_{\substack{h' \in \mathcal{A}_c \\ B_c(h') > \epsilon}} d(h, h'). \quad (11)$$

The testable question is whether a prompt increases this margin under a fixed attack or perturbation contract while retaining usefulness. When comparing different prompt contexts, use comparable admissible changes and costs; do not silently treat their continuation maps as identical.

Specify the admissible changes (for example matched donor patches, bounded activation edits, or a separate family of natural prompt perturbations) and the distance/cost d . Euclidean distance is coordinate-dependent and not automatically semantic distance. Finding a nearby unsafe state gives an *upper bound* on ρ . Failure to find one does not certify a lower bound. Finite samples, activation plots, and first-order null spaces do not prove topological separation.

6 Smallest useful pilot and stopping decisions

1. **One model, one verifiable behavioral target.** Start with unauthorized modification of protected state or a false completion report. Establish legitimate-task competence and enough actual violations before evaluating detection.
2. **Split by underlying task and mechanism.** Keep paired variants together. Fit on some mechanisms, select on development groups, and hold out a complete mechanism. Repeated calls and paraphrases are clustered observations.
3. **Compare matched monitors.** Test an individual activation probe, a small ensemble, a text monitor, and simple nuisance baselines at a shared FPR budget and recorded inference cost. Report uncertainty clustered by task base.
4. **One prompt intervention.** Measure violations, useful completion, and frozen monitor scores independently. Retain cases where readings improve but behavior does not.
5. **Target the discrepancies.** Inspect shared misses, estimate finite behavior differences, and test a small number of valid patches. Use a new held-out set after adapting the monitor to a discovered failure.

Advance when a selected monitor or additional measurement improves detection on withheld mechanisms at retained utility and matched cost. **Stop or narrow** when apparent gains come from increased false alarms, task failure, superficial cues, or test-set adaptation. Treat weak positive-event coverage as a measurement problem rather than evidence of reliable monitoring.

7 Application phrasing: keep the broad ideas, narrow the immediate claim

For a short application answer, foreground shared blind spots and held-out transfer. Keep representation geometry and prompting as methods or follow-ups within that area.

- Say “a detector that transfers across situations,” rather than “unsafe behavior that transfers.”
- Replace “exhausting all possible cases” with “testing whether probes have complementary failures on held-out violation mechanisms.”
- Replace “ideal versions of behaviors” with “features that preserve behavior-relevant distinctions across contexts.”
- Frame prompt-induced geometric separation as a question, not an assumption. The immediate test is whether prompt changes improve verified behavior or merely lower detector scores.
- Aim to identify *when* activations add value beyond strong text baselines. A negative result can identify a useful limit; improvement is not guaranteed.

References

- [1] M. MacDiarmid et al. *Simple probes can catch sleeper agents*. Anthropic Alignment Note, 23 April 2024. [Official research note](#).
- [2] R. Wang et al. *Evaluating honesty and lie detection techniques on a diverse suite of dishonest models*. Anthropic Alignment Science, 25 November 2025. [Official research article](#).
- [3] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz. *Invariant Risk Minimization*. arXiv:1907.02893, 2019. [Preprint](#).
- [4] Anthropic. *Persona vectors: Monitoring and controlling character traits in language models*. 1 August 2025. [Official research article](#).